

Research article

AOP-DRL: A deep representation learning framework for the computational prediction of antioxidant peptides

Yongzhu Zhou^{a,1}, Wanlin Liu^{b,1}, Qiao Liu^a, Jie Liu^a, Xing Yu^{a,*}, Jie Ma^{b,*}, Yunping Zhu^{b,*}^a School of Medicine, Hunan Normal University, No. 36, Lushan Road, Changsha, Hunan 410081, China^b State Key Laboratory of Medical Proteomics, Beijing Proteome Research Center, National Center for Protein Sciences (Beijing), Beijing Institute of Lifeomics, 38 Life Science Park Road, Beijing 102206, China

ARTICLE INFO

Keywords:

Antioxidant peptides
Deep representation learning
Protein language models
Text-CNN
Peptide sequence prediction

ABSTRACT

Molecular oxygen is vital for life but can generate reactive oxygen species (ROS), leading to oxidative stress and biomolecular damage. Antioxidant peptides mitigate ROS but face scalability challenges in traditional identification methods. To address this, we developed antioxidant peptides deep representation learning (AOP-DRL), a deep learning framework that combines protein language models with hierarchical convolutional networks for high-throughput prediction. The training dataset from the publicly available original study includes experimentally validated antioxidant sequences and negative controls orthogonally verified via public repositories under standard redox proteomics data partitioning protocols. Our architecture overcomes limitations in handling variable peptide lengths and capturing nonlinear residue interactions. Benchmarking experiments reveal that AOP-DRL outperforms state-of-the-art models in terms of accuracy while generalizing well across diverse datasets. Our model achieved accuracy improvements of 7.26 %, 2.57 %, 2.59 %, and 4.20 % on the prepartitioned subsets P60, P70, P80, and P90, respectively, compared with the mean accuracy of previously specialized models for antioxidant peptides. This approach provides a cost-effective, scalable alternative to wet-lab methods, accelerating the discovery of therapeutic and nutraceutical peptides. The adaptability of the framework also suggests broader applications in bioactive peptide prediction for personalized medicine and functional foods.

1. Introduction

Molecular oxygen (O₂), often termed the “molecule that sustains life” in biochemical literature, serves as a critical substrate in therapeutic interventions such as hyperbaric oxygen therapy and ischemia–reperfusion management [1]. Pathological overproduction of reactive oxygen species (ROS) through dysregulated metabolic conversion processes, particularly via mitochondrial electron transport chain leakage, initiates oxidative cascade reactions that compromise cellular homeostasis [2]. These redox-active species, which are generated primarily as unavoidable byproducts during aerobic metabolism, exhibit unpaired electron configurations that drive their electrophilic nature and transient existence. Their indiscriminate reactivity induces biomolecular modifications, including lipid peroxidation cascades, DNA strand breakage, and protein misfolding, ultimately manifesting as cumulative cellular dysfunction and senescence-associated secretory

phenotypes [3]. Recent studies have highlighted the role of ROS in cadmium-induced stress responses in liverworts [4] while also demonstrating the potential of plant-derived antioxidant peptides to mitigate such oxidative damage [5].

As multifunctional therapeutic candidates, protein-derived antioxidant peptides (AOPs) have gained prominence in redox biology research owing to their dual capacity for enzymatic and nonenzymatic antioxidant activity [6]. Unlike small-molecule antioxidants such as ascorbic acid, AOPs demonstrate superior membrane permeability and multi-target modulation capabilities through amphipathic structural domains [7]. Their therapeutic potential extends to mitigating the pathogenesis of age-related disorders, including oncogenesis through/pathway regulation, atherosclerotic plaque stabilization via oxidation inhibition, and neuroprotection against beta-amyloid aggregation in Alzheimer's disease [8–10]. In food technology applications, AOPs serve as natural preservatives through synergistic inhibition of lipid radical propagation

* Corresponding authors.

E-mail addresses: xingyu@hunnu.edu.cn (X. Yu), majie@ncpsb.org.cn (J. Ma), zhuyunping@ncpsb.org.cn (Y. Zhu).¹ These authors contributed equally to this work.

cycles, effectively preserving organoleptic properties in meat products and preventing oxidative rancidity in dairy matrices [11,12].

Nevertheless, systematic identification and functional characterization of antioxidant peptides remain formidable challenges in biochemical research. Conventional experimental workflows encompassing isobaric tags for relative and absolute quantitation (iTRAQ) and 2D-PAGE face inherent limitations in temporal efficiency and economic feasibility, particularly when scaling for high-throughput screening [13]. The convergence of transformer-based architectures and self-supervised learning paradigms has revolutionized computational approaches for *de novo* antioxidant peptide discovery.

The evolution of bioactive prediction models has demonstrated a clear epistemological shift from single-modality pattern recognition to context-aware multimodal integration, driven by successive advancements in feature representation learning. The developmental trajectory began with unimodal architectures specialized in single molecular descriptors. ChemBERTa, which implements molecular-level feature learning via transformer encoders, pioneered transformer-based representation learning, where stacked self-attention layers pretrained via masked language modeling objectives decode SMILES notation sequences into substructure-aware embeddings [14]. AnOxPePred uses hierarchical convolutional neural networks (CNNs) with residual connections to address this gap, explicitly modeling both short-range amino acid interactions and long-distance physicochemical dependencies within linear peptide chains [15]. The field then progressed toward evolutionary-scale pretraining to capture phylogenetic constraints. Evolutionary sequence model 2 (ESM-2) uses evolutionary-scale sequence embeddings to generate context-sensitive embeddings that encode structural fold propensities critical for functional peptide prediction [16]. Unlike conventional approaches, ESM-2 implements position-aware attention layers that dynamically adjust receptive fields, enabling effective processing of polypeptides exceeding 100 residues while maintaining computational efficiency. In addition, next-generation models have embraced cross-modal fusion architectures to overcome information silos. UniDL4BioPep integrates multimodal representations, dynamically weighting physicochemical attributes against evolutionary contexts for enhanced bioactivity inference [17]. DGAOP combines quantum-chemical calculations with deep learning, enabling co-optimization of redox potentials and structural stability through differentiable molecular dynamics [18]. Moreover, protein sequence pretraining models such as BERT-BFD [19] and OntoProtein [20] also hold potential for application in AOPs. BERT-BFD employs masked language modeling objectives trained on the Big Fantastic Database (BFD), a comprehensive corpus containing 2.122 billion evolutionarily diverse protein sequences that were originally curated [19,21]. OntoProtein integrates Gene Ontology annotations with graph neural networks, establishing explicit connections between sequence motifs and functional classifications within biological systems [20]. These computational frameworks employ divergent representation learning strategies to map peptide sequences into high-dimensional latent spaces, enabling the discovery of nonlinear structure–activity relationships through advanced neural architectures. In addition to deep learning-based architectures, practical AOP prediction platforms such as AnOxPP [22] and iAnOxPep [23] have shown promising performance by integrating handcrafted features, amino acid descriptors, and ensemble learning techniques, offering effective alternatives in real-world applications.

Although existing computational predictors have advanced the field, three fundamental limitations persist: sequence length adaptability, nonlinear dependency modeling, and biochemical knowledge integration. The global attention mechanisms of ChemBERTa demonstrate suboptimal resolution in capturing residue-level interactions, particularly in short-range spatial configurations. Although effective for molecular contexts, this approach ignored peptide-specific biophysical patterns, which prompted the emergence of adaptive multiscale frameworks. ESM-2 generates high-dimensional sequence embeddings;

however, its static parameterization limits dynamic adaptation to context-dependent conformational changes in peptide structures. Fixed-receptive-field architectures in AnOxPePred and UniDL4BioPep exhibit exponentially decaying gradient propagation efficiency when processing sequences exceeding 30 residues.

To address these limitations, we propose the antioxidant peptides deep representation learning framework (AOP-DRL), which is a dual-stream architecture that synergistically integrates the evolutionary-scale peptide embeddings of ESM-2 with the multiscale feature abstraction capabilities of the text convolutional neural network (Text-CNN) through residual concatenation mechanisms. The Text-CNN module employs dilated convolutional layers with adaptive receptive fields to progressively capture hierarchical sequence patterns, whereas attention-guided pooling operations preserve critical positional information during feature dimensionality reduction. The framework further incorporates semisupervised contrastive learning, leverages unlabeled data to enhance latent space regularization, and improves generalizability under limited conditions. In summary, AOP-DRL presents a systematic solution to key challenges in antioxidant peptide identification by combining deep semantic modeling with effective sequence feature extraction. Through end-to-end representation learning and task-specific adaptation, AOP-DRL achieves superior performance across multiple evaluation metrics, offering a promising approach for computational peptide characterization.

2. Methods

2.1. Dataset

The benchmark dataset was curated from the original AnOxPePred study [15], following strict data partitioning protocols established in redox proteomics research. The curated dataset comprises 687 experimentally validated antioxidant sequences (456 radical scavengers and 231 metal chelators) and 717 negative control peptides, with all entries verified through orthogonal validation in publicly accessible repositories. The multidimensional characteristics of the antioxidant peptides were elucidated through integrated triaxial analysis (Fig. 1). Through complementary dimensions (length–sequence–composition), these panels collectively establish a three-dimensional analytical framework, thereby providing a data-driven scaffold for subsequent functional investigations.

The hydroxyl radical ($\bullet\text{OH}$) and superoxide anion ($\text{O}_2^{\bullet-}$) scavenging capacities were selected as validation benchmark indicators because of their established correlation with cellular oxidative damage pathways. The experimental design utilized prepartitioned subsets (P60–P90) from the original study, representing progressive sequence identity thresholds from 60 % to 90 % through 5-fold cross-validation. This stratified partitioning strategy guarantees interfold sequence identity below designated thresholds, thereby enhancing model generalizability across sequence similarity regimes while ensuring evaluation reliability.

3. Model construction

3.1. Peptide representation

The tokenization schema implemented in our framework comprises 26 discrete tokens designed to comprehensively represent peptide sequence diversity. This lexicon contains five functional control tokens ([PAD], [UNK], [CLS], [SEP], and [MASK]) following BERT-style pre-training conventions; canonical representations for the 20 proteinogenic amino acids; and a dedicated token for noncanonical residues. Notably, the tokens were strategically incorporated to encapsulate selenocysteine, pyrrolysine, and post-translational modifications, which were aligned with residue annotations in the Protein Data Bank (PDB) and UniProt Knowledgebase (UniProtKB) reference databases. This hierarchical tokenization architecture enhances the representational capacity

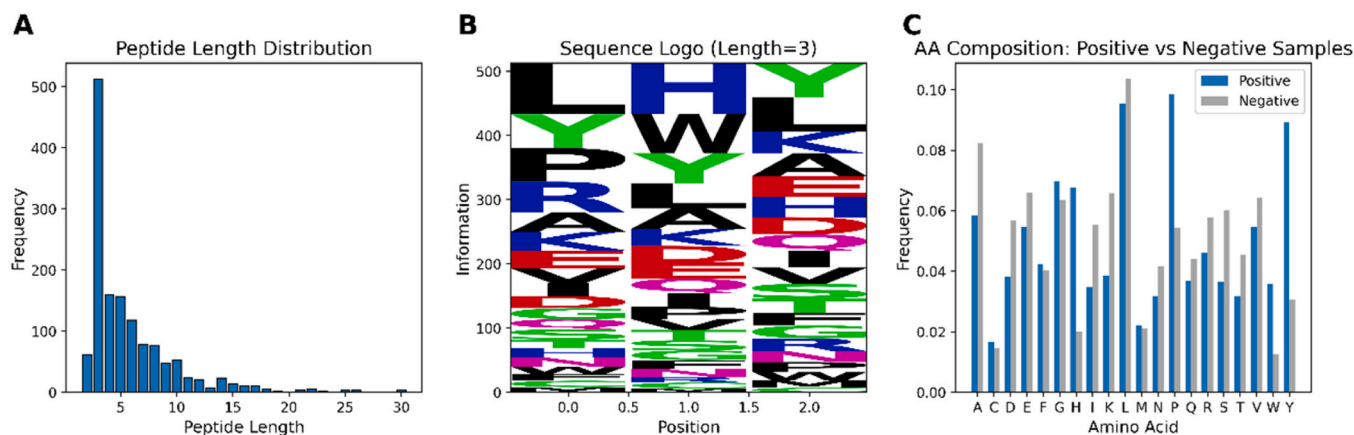


Fig. 1. Multidimensional characterization of peptide features in the AOP dataset. (A) Peptide length distribution. Tripeptides show peak abundance (3–15 residue segments account for > 80 % of the total); the oligopeptide frequency decreases exponentially with increasing chain length (2–30 residues). (B) Sequence logo (length = 3). Position-specific amino acid bias revealed by information entropy analysis of tripeptides; color-coded logos denote conserved residues at N-terminal, central, and C-terminal positions. (C) AA composition. Positive vs. negative samples: Differential abundance analysis identifies characteristic amino acids distinguishing positive and negative sample groups.

for rare biochemical variants while maintaining backward compatibility with standard sequence analysis pipelines through adaptive token embedding layers. This enabled the system to demonstrate enhanced capability in interpreting atypical peptide components while maintaining compatibility with existing analytical tools owing to its intelligent and adaptable translation layers. The sequence transduction pipeline converts raw amino acid strings into numerical tensor representations via a learnable embedding layer, enabling end-to-end differentiable operations that preserve spatial and chemical relationships during model optimization. This process converts amino acid sequences into numerical representations via an adaptive layer that improves as the system learns, enabling the model to better understand peptides by preserving amino acid order and key properties during training.

3.2. Selection and optimization of the pretraining model

The selection of an appropriate pretraining paradigm constitutes a critical methodological consideration in computational proteomics. Contemporary computational biology research has established that protein sequences exhibit intrinsic linguistic properties analogous to natural language syntax, where amino acid arrangements convey functional semantics through spatial organization [24]. This conceptual framework has enabled the adaptation of transformer-based natural language processing (NLP) architectures for deciphering complex patterns in amino acid sequences through self-supervised learning methodologies [25].

Among contemporary implementations, BERT-BFD exemplifies the classical bidirectional encoder architecture, whereas OntoProtein pioneers knowledge-enhanced representation learning through ontological constraints. The architectural innovation of ESM-2 facilitates comprehensive context modeling through hierarchical feature extraction, which is particularly valuable for resolving long-range allosteric interactions and tertiary structure dependencies in protein engineering applications. In this study, our protocol evaluated these three pretrained language models to assess their suitability for peptide bioactivity prediction. The dataset was partitioned into an 80:20 stratified split between training and test sets, ensuring proportional representation of functional classes across both subsets.

3.3. Design and implementation of the fine-tuning layer

Pretraining models such as ESM-2 can effectively capture the global features of peptide sequences in the pretraining stage; however, their general feature extraction ability has limited adaptability to specific

tasks. Therefore, the introduction of the fine-tuning layer is crucial. The fine-tuning layer can optimize the feature representation on the basis of the pretrained model and improve the adaptability of the model to specific peptide sequence classification tasks [26,27].

In the fine-tuning phase, we evaluated four different fine-tuning layers: the fully connected layer (FC), Text-CNN [28], long short-term memory (LSTM) [29], and bidirectional long short-term memory (BLSTM) [30]. Among them, Text-CNN has been widely used in various sequence data processing tasks [31–33]. Text-CNN extracts local features through a convolution operation, which can effectively capture the patterns and local dependencies in peptide sequences and is especially suitable for processing short-range features and local sequence patterns in antioxidant peptide sequence data.

To implement the Text-CNN fine-tuning layer, we first utilized the last hidden layer output of the pretrained ESM-2 model as the input feature representation. Specifically, for each input sequence, the final hidden states were extracted and passed to a convolutional encoder composed of a single convolutional layer with a kernel size of 3 and 64 filters. To reduce overfitting and enhance generalization, a dropout layer with a rate of 0.2 was applied. Max-pooling was then used to extract the most informative local features from the convolutional outputs. The pooled features were concatenated and passed through a fully connected layer for binary classification. Unlike traditional RNN-based architectures, this design enables parallel processing of short-range patterns and reduces training time while maintaining high accuracy.

Throughout training, both the pretrained backbone and fine-tuning layer were updated via backpropagation using the Adam optimizer. We did not alter the structure or hyperparameters of the ESM-2 encoder, ensuring that the performance improvements originated solely from the added task-specific fine-tuning layer. This modular design allowed the model to retain the generalization capacity of ESM-2 while improving its performance on the classification task.

Text-CNN extracts local features through convolution operations to capture patterns and local dependencies in peptide sequences. Unlike traditional recurrent neural networks (RNNs) or LSTM, Text-CNN extracts feature information at different scales in parallel through multiple convolution kernels. Text-CNN performs dimensionality reduction and feature extraction through the pooling layer.

The calculation process of Text-CNN can be summarized in the following steps:

1. Convolution operation: The input peptide sequence is convoluted by multiple convolution kernels to extract local features. The input sequence is assumed to be $X = [x_1, x_2, \dots, x_n]$, where W_j is the convolution kernel, b_j is the bias term, $*$ represents the convolution operation, and h_j

is the feature after convolution. The calculation process of the convolution operation is as follows:

$$h_j = \text{ReLU}(W_j * X + b_j) \quad (1)$$

2. Activation function: The output after the convolution operation is nonlinearly transformed by the activation function (ReLU) to enhance the expression ability of the model:

$$h_j = \text{ReLU}(h_j) \quad (2)$$

3. Pooling operation: The pooling layer is used to reduce the dimension of the convolutional features and retain the most significant feature information. Among them, p_j is the feature after pooling, and the calculation of the maximum pooling operation is:

$$p_j = \max(h_j) \quad (3)$$

4. Fully connected layer: The pooled features are combined through the fully connected layer to generate the final feature representation. Assuming that the pooled feature vector is $p = [p_1, p_2, \dots, p_m]$, where W_f is the weight of the fully connected layer, b_f is the bias term, and y is the output of the fully connected layer, the calculation process is:

$$y = W_f \cdot p + b_f \quad (4)$$

5. Softmax classifier: According to the set threshold (0.5), the Softmax classifier performs classification prediction. If $P(\text{AOP}) > 0.5$, the sequence is classified as an antioxidant peptide (AOP); otherwise, the sequence is classified as a nonantioxidant peptide (non-AOP). The calculation formula of the Softmax function is:

$$P(\text{AOP}) = \frac{\exp(y_{\text{AOP}})}{\sum_k \exp(y_k)} \quad (5)$$

As shown in [Supplementary Figure 1](#), our classification framework processes ESM-2 embeddings through three sequential computational stages:

1. **Feature Abstraction:** The pretrained embeddings undergo multi-scale convolutional filtering via Text-CNN to extract hierarchical sequence motifs.
2. **Feature Integration:** A dense layer with nonlinear activation combines high-order feature representations.
3. **Discriminative Classification:** The classifier generates probabilistic predictions for antioxidant/nonantioxidant peptide categorization.

This hierarchical architecture enables robust discrimination between bioactive antioxidant peptides and nonantioxidant sequences through systematic transformation of raw embeddings into discriminative feature space.

3.4. Performance evaluation

The predictive performance of our framework was rigorously assessed via five established evaluation metrics: accuracy (ACC), the Matthews correlation coefficient (MCC), recall, the F1 score, and the area under the ROC curve (AUC). The formal definitions are as follows:

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FN) \times (TP + FP) \times (TN + FP) \times (TN + FN)}} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

$$\text{F1} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (9)$$

$$\text{AUC} = \int_0^1 \text{TPR}(x) d\text{FPR}(x) \quad (10)$$

To assess the statistical significance of performance differences among models, we conducted normality tests on the prediction results obtained through five-fold cross-validation. Based on the test outcomes, we selected appropriate statistical methods: the Student's *t*-test was used for metrics that satisfied the assumptions of normality and homogeneity of variance, while the Wilcoxon rank-sum test was applied for those that did not. All statistical analyses were performed on the cross-validation results to ensure the reliability of comparative evaluations.

4. Results

The computational experiments were conducted on a dedicated node within the GPU cluster at the National Center for Protein Sciences (Beijing), utilizing dual NVIDIA Tesla V100 GPUs. All the implementations utilized the PyTorch 2.2 framework with CUDA 11.8 acceleration. A linear warmup strategy with subsequent decay was applied to the learning rate schedule during the 20-epoch training phase. To ensure methodological rigor, comprehensive benchmarking included comparative analysis of pretrained architectures and tuning layers, with AOP-DRL performance rigorously validated against state-of-the-art baselines through systematic evaluation protocols.

4.1. Framework of AOP-DRL

Accurately predicting peptide bioactivity requires balancing the acquisition of generalizable sequence representations with the retention of interpretable local motifs critical for redox functions. The AOP-DRL framework addresses this through a streamlined architecture ([Fig. 2](#)) that combines evolutionary-scale pretraining with targeted feature refinement. We first utilize the pretrained ESM-2 model (650 M parameters), which produces residue-level contextual embeddings of size 1280 for each amino acid token. To adapt these general protein language representations to antioxidant peptide prediction, we perform domain-specific fine-tuning on a curated dataset with 1404 peptides sequences in total, including 687 experimentally validated antioxidant peptides and 717 antioxidant-negative peptides. During this fine-tuning stage, we freeze the lower transformer layers of ESM-2 and update only the higher layers over 20 epochs using the Adam optimizer with a learning rate warm-up and decay schedule. This fine-tuning procedure adapts the pretrained model to our specific task.

The resulting $L \times 1280$ residue-level embeddings preserve broad phylogenetic and biochemical context while enabling downstream local motif analysis. A specialized Text-CNN module with a kernel size of three residues processes these embeddings through 64 parallel filters, capturing spatially localized motifs associated with antioxidant activity. To further enhance feature relevance and support biological discrimination, an attention mechanism is applied over the convolutional outputs. This component adaptively reweights different regions of the peptide sequence by assessing the contextual importance of detected motifs, effectively linking local residue patterns with global sequence understanding. The interaction between CNN-extracted motifs and attention-based weighting enables the model to prioritize biologically informative substructures during classification. This synergy between localized pattern recognition and global context modeling helps bridge the gap between predictive performance and mechanistic interpretability in peptide function prediction.

To mitigate overfitting within the lightweight architecture, a dropout rate of 20 % is applied to the aggregated CNN features before final classification via a fully connected layer. Through this biologically motivated modular design, AOP-DRL achieves a strong balance between efficiency, accuracy, and interpretability in modeling antioxidant peptide functionality.

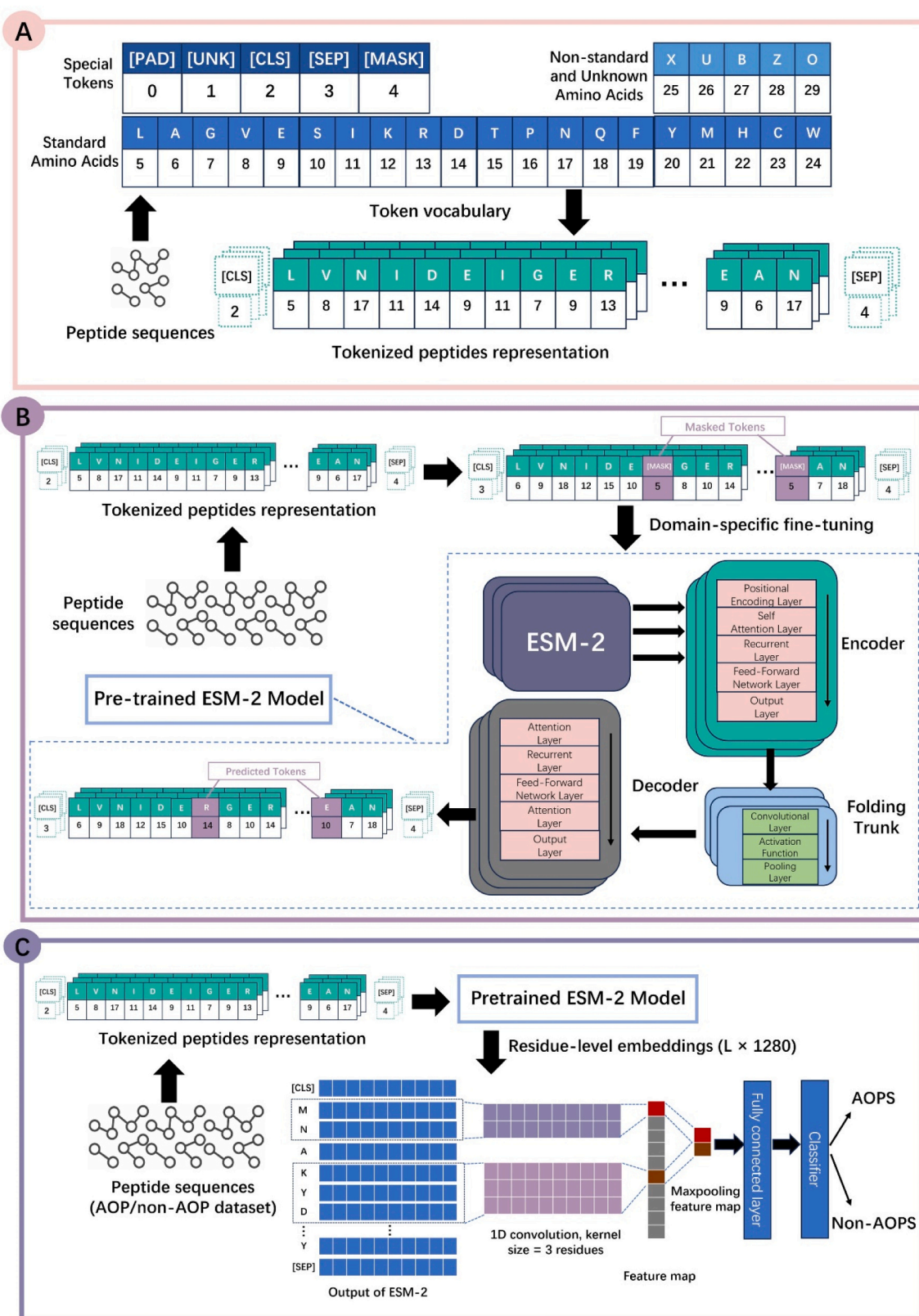


Fig. 2. Framework of AOP-DRL. (A) Peptide representation: Input amino acid sequences are tokenized and encoded using the pretrained ESM-2 model, which produces residue-level contextual embeddings of size 1280 for each amino acid token ($L \times 1280$ matrix, L represents the peptide length). (B) Domain adaptation: The pretrained ESM-2 model is fine-tuned on a curated antioxidant peptide dataset, including 687 experimentally validated positive sequences and 717 negative sequences, to adapt general protein language representations for this task. This step selectively updates higher transformer layers using the Adam optimizer with a warm-up and decay schedule. (C) Fine-tuning layer: The residue-level embeddings are processed by a Text-CNN model with kernel size of three residues to capture tri-peptide motifs, followed by attention-based aggregation and final classification into antioxidant or non-antioxidant peptides. This modular design enables systematic transformation from raw sequences to functional predictions while maintaining biological interpretability.

4.2. Evaluation of different pretraining models

To rigorously evaluate the comparative performance of pretrained architectures in peptide language modeling tasks, we conducted benchmark evaluations of the BERT-BFD, OntoProtein, and ESM-2 models under identical experimental conditions. All the models employed a unified tokenization schema for sequence encoding, with encoded representations subsequently processed through a fully connected classification layer.

As illustrated in Fig. 3, the BERT-BFD model demonstrated suboptimal performance in critical evaluation metrics (AUC 0.791, F1 75.19 %, MCC 0.53). In contrast, ESM-2 exhibited superior predictive capabilities across multiple dimensions of model assessment (AUC 0.847, F1 77.61 %, MCC 0.57). Although OntoProtein achieved commendable results (AUC 0.780, F1 70.78 %, MCC 0.40), its performance marginally underperformed relative to the state-of-the-art ESM-2 architecture.

A comprehensive analysis of the classification performance (Fig. 4) further substantiates these findings. The ROC curve analysis (Panel A) revealed the superior discriminative capacity of ESM-2 (AUC = 0.85), representing 7.6 % and 8.2 % improvements over BERT-BFD (AUC = 0.79) and OntoProtein (AUC = 0.78), respectively. Quantitative analysis of the PR curves (Panel B) demonstrated the superior overall performance (AUC = 0.84) of ESM-2 compared with those of BERT-BFD (AUC = 0.79) and OntoProtein (AUC = 0.80). The curve morphology reveals that ESM-2 maintains higher precision values across all recall thresholds, particularly exhibiting shallower performance degradation in high-recall regimes (Recall > 0.7) relative to competing models. Score distribution (Panel C) analysis revealed a distinct bimodal pattern in the ESM-2 predictions, indicating sharper decision boundaries. Finally, the confusion matrix (Panel D) quantifies the operational robustness of ESM-2, achieving 78.2 % precision (104/133) and 77.0 % recall (104/135) with balanced error rates (FP = 29, FN = 31).

These findings collectively suggest that the advanced sequence modeling paradigm of ESM-2 offers enhanced adaptability when processing biologically diverse peptide sequences characterized by complex structural hierarchies.

4.3. Fine-tuning layer evaluation

We then systematically evaluated four neural architectures for

optimizing pretrained peptide embeddings across five performance metrics. Our methodology employs pretrained amino acid embeddings for feature extraction from peptide sequences, subsequently applying distinct fine-tuning architectures for classification optimization. Four prevalent neural architectures were comparatively evaluated: FC, Text-CNN, LSTM, and BLSTM. The FC baseline exclusively utilizes the [CLS] token embeddings for classification, whereas the other architectures leverage full-sequence encodings for hierarchical feature learning.

The benchmark evaluations (Fig. 5) revealed the following results: FC (ACC = 84.37 %, MCC = 0.54), LSTM/BLSTM (77.94 %/77.22 %, 0.56/0.54), and Text-CNN (79.36 %, +2.14 % vs. BLSTM; MCC = 0.59, F1 = 78.20 %, AUC = 0.852). These metrics validate the multiscale convolutional filters of Text-CNN as optimal for capturing spatially conserved motifs, whereas temporal models (LSTM/BLSTM) underperformed despite sequence-order sensitivity.

The integration of Text-CNN with ESM-2 embeddings yields enhanced predictive capability relative to the baseline ESM-2 model (ACC: 79.36 % vs. 78.65 %), corroborating recent findings in structural bioinformatics [26,27]. This performance enhancement underscores the complementary nature of global evolutionary patterns captured by protein language models and localized functional motifs identified through task-specific convolutional operations. The hierarchical architecture effectively synergizes the global sequence context awareness of ESM-2 with the capacity for detecting short-range physicochemical interaction patterns of Text-CNN, thereby optimizing antioxidant peptide classification through multimodal feature integration.

4.4. Ablation experiment

To systematically evaluate the contribution of each component within the AOP-DRL framework, we conducted a comprehensive set of ablation experiments that targeted both the pretrained encoder and the downstream fine-tuning module. First, to investigate the role of pretrained representations, we removed the ESM-2 encoder entirely and directly applied the Text-CNN classifier to one-hot encoded peptide sequences, forming the AOP-DRL (Without Pretraining) variant. This configuration resulted in a substantial performance drop (ACC: 0.5552, F1: 0.2331, AUC: 0.6668, MCC: 0.13), confirming that pretrained protein language models are essential for capturing high-level biochemical semantics and ensuring prediction robustness.

To further isolate the influence of the encoder architecture and

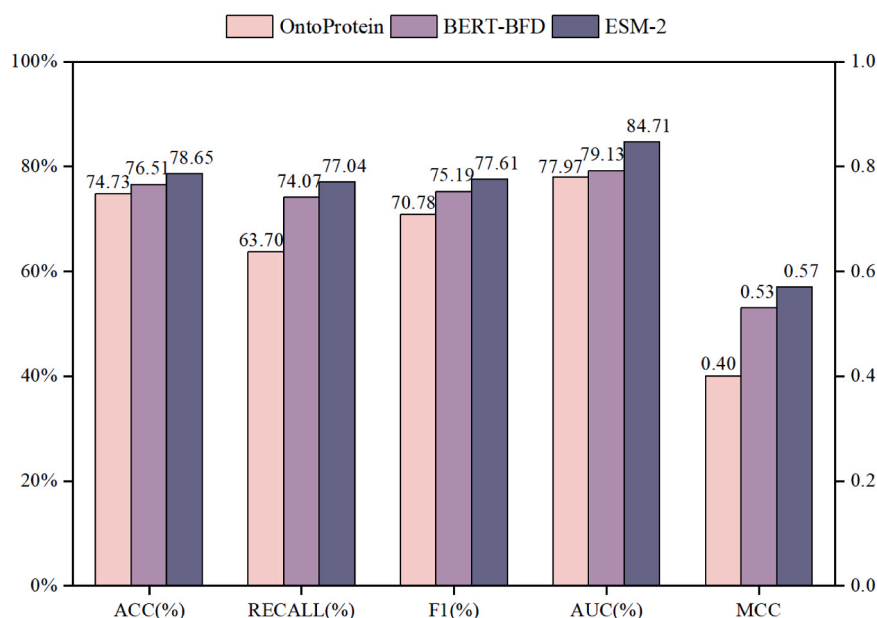


Fig. 3. Evaluation of different pretrained models. The ACC, RECALL, F1, AUC and MCC of OntoProtein, BERT-BFD and ESM-2 are presented.

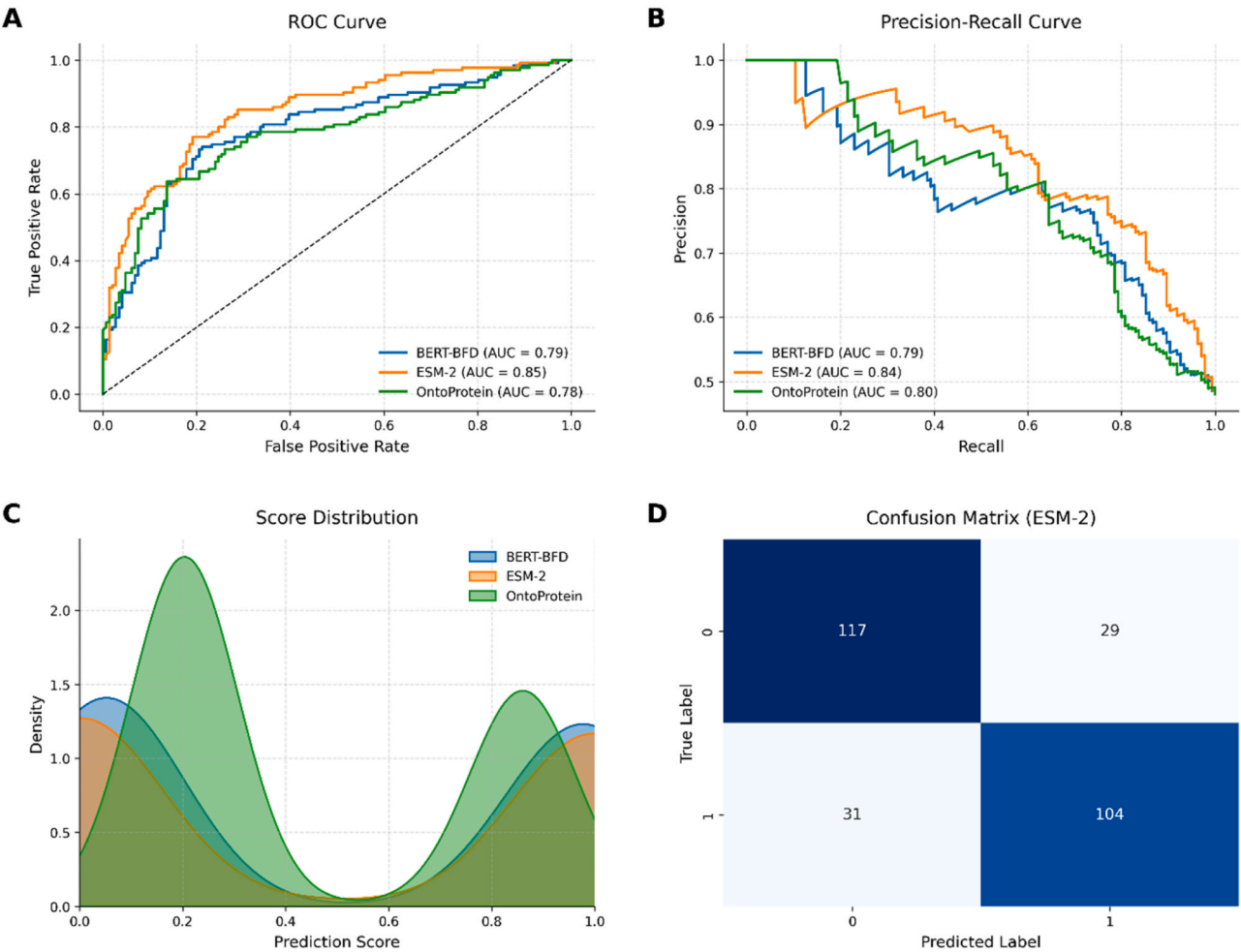


Fig. 4. Comparative evaluation of protein language models. (A) ROC curves of the models. ESM-2 (AUC = 0.85) demonstrates superior discriminative capacity compared with BERT-BFD (0.79) and OntoProtein (0.78), with the diagonal dashed line representing random classification. (B) Precision–recall curves of the models. ESM-2 maintains AUC = 0.84 performance, showing enhanced precision retention at elevated recall levels (> 0.7) relative to that of the comparator models. (C) Score distribution of the models. Pronounced bimodal separation in the ESM-2 score distribution contrasts with overlapping confidence profiles in BERT-BFD and OntoProtein. (D) Confusion matrix of ESM-2. ESM-2 achieves balanced predictive accuracy (precision = 78.2 %, recall = 77.0 %) on the standardized test cohort (n = 281).

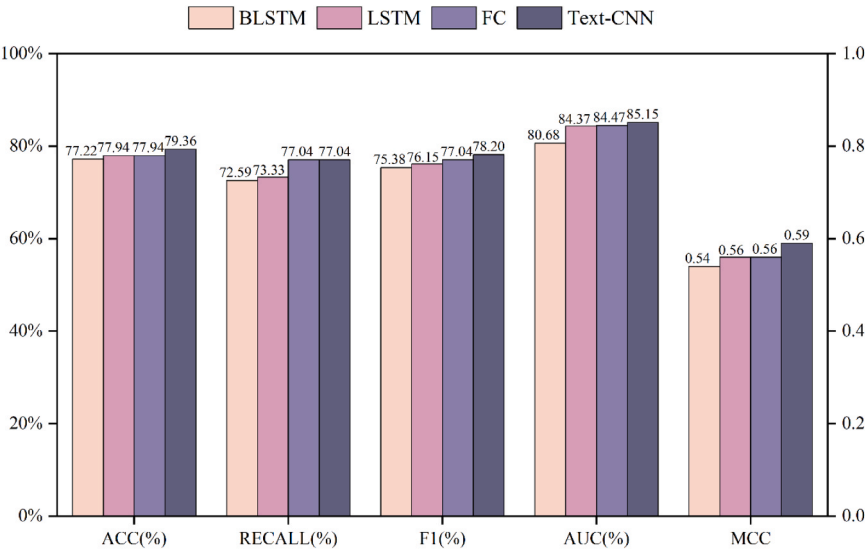


Fig. 5. Performance of different fine-tuning layers. The ACC, RECALL, F1, AUC and MCC of the BLSTM, LSTM, FC and Text-CNN are presented.

pretraining strategy, we replaced ESM-2 with BERT-BFD, a comparatively weaker transformer-based model trained on a less comprehensive corpus. This variant, referred to as AOP-DRL (Weak Pretrained Model), retained the same Text-CNN refinement layer and achieved moderate performance (ACC: 0.7580, F1: 0.7302, AUC: 0.7688, MCC: 0.52), highlighting the importance of both large-scale pretraining and model expressiveness for downstream peptide classification. We also evaluated a variant using frozen ESM-2 representations without any fine-tuning, AOP-DRL (Without Fine-tuning). This model outperformed both the no-pretraining and weak pretraining configurations (ACC: 0.7865, F1: 0.7761, AUC: 0.8471, MCC: 0.57), indicating that the contextual embeddings learned by ESM-2 are inherently effective even without task-specific adaptation.

The full AOP-DRL model, which combines ESM-2 with a Text-CNN fine-tuning layer, consistently achieved the best results (ACC: 0.7936, F1: 0.7820, AUC: 0.8515, MCC: 0.59), demonstrating the complementary synergy between global contextual representations and localized motif detection. The comparative results of the ablation variants are summarized in [Supplementary Figure 2](#).

To assess the generalizability of these findings, we extended the ablation experiments to four benchmark datasets (P60, P70, P80, and P90), using five-fold cross-validation for each. Across all the datasets, the full AOP-DRL model consistently outperformed its ablated counterparts. For example, on the P60 dataset, AOP-DRL achieved an accuracy of 69.17 %, an F1 score of 64.22 %, and an MCC of 0.38, surpassing those of AOP-DRL (Without Fine-tuning) (67.57 %, 62.09 %, 0.35) and AOP-DRL (Weak Pretrained Model) (65.29 %, 60.20 %, 0.30). The gap was particularly pronounced compared with AOP-DRL (Without Pre-training), which yielded only 58.62 % accuracy and a 22.14 % F1 score.

These results further confirm the contribution of each module. The performance difference between AOP-DRL and the ESM-2-only variant highlights the importance of the Text-CNN layer in refining pretrained embeddings, whereas the inferior results of the weakly pretrained BERT-BFD variant underscore the decisive impact of the encoder architecture and pretraining corpus quality. The Text-CNN-only baseline performed the worst across all datasets, especially in terms of recall and F1 score, suggesting that local filters alone are insufficient to capture the complex biochemical patterns associated with antioxidant activity. The comparative performance of all the variants across the four datasets is illustrated in [Supplementary Figure 3](#).

4.5. Comparison with existing methods

To rigorously assess the comparative efficacy of various fine-tuning architectures against established methodologies, we implemented a stratified five-fold cross-validation protocol across multithreshold datasets (P60–P90). The results of the four baseline methods were all obtained through independent reproducibility experiments. We retrieved the publicly available source codes and execution environments from the respective publications and strictly followed the original parameter settings provided by the authors. No modifications were

made to the model structures or hyperparameters to ensure that the reproduced results were consistent with the original methods and thus comparable. In cases where multiple versions or configuration files were available, we adopted the standard version explicitly recommended by the original authors.

As documented in [Tables 1–4](#), although all the evaluated models exhibited methodologically consistent performance variance across data partitions, the AOP-DRL architecture consistently demonstrated comprehensively improved overall performance. Quantitative analysis revealed that AOP-DRL tends to achieve favorable performance in terms of ACC and MCC across all the homology thresholds. For example, AOP-DRL attained 76.21 % ACC on the P90 dataset, which represents a modest improvement of approximately 1.5 % over the next best-performing model. Although the FC model achieved a slightly higher F1 score (73.04 % vs. 72.74 %) on the P70 dataset and the BLSTM showed a minor recall advantage (75.74 % vs. 73.34 %) on the P90 dataset, the Text-CNN backbone within AOP-DRL generally yielded higher AUC values (up to 82.22 %) and a consistent edge in MCC-ACC composite metrics, with an average improvement of approximately 3.2 %. Our model achieved ACC improvements of 7.26 %, 2.57 %, 2.59 %, and 4.20 % on the prepartitioned subsets P60, P70, P80, and P90, respectively, compared with the mean ACC of previously specialized models for antioxidant peptides. Although these gains are relatively moderate, they indicate that AOP-DRL performs robustly and maintains stable results across diverse evaluation criteria. The combined MCC–ACC analysis suggests balanced and reliable predictive behavior, particularly in addressing challenges posed by sequence length variability and class imbalance in peptide datasets.

As further validated in [Fig. 6](#), the radar chart visually consolidates the statistical dominance of AOP-DRL over general-purpose and specialized models across multithreshold datasets (P60–P90). The radial symmetry of the performance profile of AOP-DRL—consistently enveloping competitor models in all the metrics—objectively corroborates its comprehensive superiority, which aligns with the cross-validation outcomes in [Tables 1–4](#). Statistical significance tests further confirm that these performance differences are not due to random variation ([Supplementary Figures 4–7](#)). This visual evidence reinforces its adaptability to data heterogeneity and metric-balanced efficacy.

These empirical findings not only validate the superior feature extraction capabilities of our framework through its hierarchical convolutional operations but also highlight its decision boundary stability advantages over conventional recurrent networks and linear classifiers in complex bioinformatics applications.

4.6. AOP-DRL web server

To facilitate the practical application of antioxidant peptide prediction, we developed an online platform called the AOP-DRL Web Server, which provides researchers with a convenient and efficient way to analyze peptide and protein sequences. The front end of the server is built via Vue 3, while the algorithmic core is implemented in Python and

Table 1
Performance comparison of AOP prediction models on the P60 dataset.

Classification	Method	Metrics				
		ACC (%)	RECALL (%)	F1 (%)	AUC (%)	MCC
General-purpose models	AOP-DRL	69.17	59.90	64.22	73.10	0.38
	BLSTM	64.11	58.30	59.58	71.10	0.28
	LSTM	64.38	51.37	56.43	68.32	0.29
	FC	61.45	54.22	56.05	67.51	0.23
	ESM-2	67.57	58.45	62.09	71.13	0.35
Specialized models for AOPs	AnOxPePred-FRS	63.87	68.06	62.09	67.75	0.32
	UniDL4BioPep	63.32	60.34	59.54	71.21	0.27
	DGAOP	64.68	50.96	55.77	71.86	0.30
	ChemBERTa	66.10	49.16	56.26	71.03	0.30

Note: Bold values represent the best performance for each metric.

Table 2
Performance comparison of AOP prediction models on the P70 dataset.

Classification	Method	Metrics				
		ACC (%)	RECALL (%)	F1 (%)	AUC (%)	MCC
Our Model General-purpose models	AOP-DRL	74.93	70.20	72.83	80.52	0.50
	BLSTM	74.22	69.72	72.19	79.36	0.48
	LSTM	73.36	68.11	70.96	80.33	0.46
	FC	74.57	72.26	73.04	80.39	0.49
	ESM–2	74.07	73.24	73.01	80.65	0.48
Specialized models for AOPs	AnOxPePred-FRS	74.13	68.77	71.75	79.03	0.48
	UniDL4BioPep	71.94	73.58	71.48	79.97	0.44
	DGAOP	73.22	73.69	72.50	80.16	0.46
	ChemBERTa	72.93	64.94	69.66	80.14	0.46

Note: Bold values represent the best performance for each metric.

Table 3
Performance comparison of AOP prediction models on the P80 dataset.

Classification	Method	Metrics				
		ACC (%)	RECALL (%)	F1 (%)	AUC (%)	MCC
Our Model General-purpose models	AOP-DRL	75.14	73.28	73.88	81.69	0.50
	BLSTM	74.00	73.45	72.99	80.59	0.48
	LSTM	74.93	73.34	73.73	80.67	0.50
	FC	73.08	70.66	71.57	80.90	0.46
	ESM–2	73.93	72.34	72.72	80.81	0.48
Specialized models for AOPs	AnOxPePred-FRS	74.32	65.68	70.97	79.11	0.49
	UniDL4BioPep	73.01	68.07	70.79	80.73	0.46
	DGAOP	73.50	75.11	73.15	80.38	0.47
	ChemBERTa	72.15	65.90	69.34	80.29	0.45

Note: Bold values represent the best performance for each metric.

Table 4
Performance comparison of AOP prediction models on the P90 dataset.

Classification	Method	Metrics				
		ACC (%)	RECALL (%)	F1 (%)	AUC (%)	MCC
Our Model General-purpose models	AOP-DRL	76.21	73.96	74.94	82.22	0.52
	BLSTM	74.93	75.74	74.46	81.73	0.50
	LSTM	74.64	72.93	73.45	81.82	0.49
	FC	73.01	72.48	72.12	80.55	0.46
	ESM–2	75.07	72.34	73.65	81.67	0.50
Specialized models for AOPs	AnOxPePred-FRS	74.25	67.12	71.42	79.00	0.49
	UniDL4BioPep	72.22	68.63	70.31	80.15	0.44
	DGAOP	73.43	70.42	71.87	81.09	0.47
	ChemBERTa	72.65	67.01	70.20	80.33	0.45

Note: Bold values represent the best performance for each metric.

served via FastAPI, ensuring smooth interaction between users and the prediction model. The server allows users to submit single or multiple peptide sequences (up to 500), either by pasting them directly or uploading “.txt” or “.fasta” files. Upon submission, the system evaluates the antioxidant potential of each peptide via the AOP-DRL model and generates a downloadable report containing prediction probabilities and related physicochemical properties. In addition, the platform supports protein-level analysis by enabling users to upload a protein sequence (up to 500 amino acids) and choose enzymatic digestion methods, including trypsin, pepsin, and chymotrypsin, individually or in combination. The server performs in silico digestion, recommends an optimal cleavage order, and predicts the antioxidant activity of the resulting peptides. To assist users, a dedicated HELP page is also provided with clear usage instructions and examples. The AOP-DRL Web Server is freely available and designed to support antioxidant peptide discovery through a combination of deep learning techniques and accessible web-based tools. Fig. 7 illustrates the workflow of the web server, which can be accessed at <http://aop-drl.ncpsb.org.cn>.

5. Discussion

In this study, we developed AOP-DRL, an antioxidant peptides deep representation learning framework for the prediction of antioxidant peptides. This investigation systematically evaluated the impact of pretraining architectures and fine-tuning strategies on antioxidant peptide classification performance. Compared with the BERT-BFD and OntoProtein models, the ESM-2 model has superior predictive ability, which is attributable to two principal technical advantages. First, the 34-layer transformer architecture of ESM-2 with 650 million learnable parameters enables hierarchical modeling of complex sequence–structure relationships through multihead attention mechanisms. Second, its pretraining leverages the curated collection of 50 million evolutionarily diverse protein sequences in the UniRef50 database, providing comprehensive coverage of conserved functional domains. In contrast, although the training corpus of BERT-BFD (BFD database) contains 400 million sequences, its lack of stringent quality control introduces noise that compromises evolutionary signal detection. Although innovative, the cross-modal knowledge integration of OntoProtein is less effective for capturing fundamental biophysical peptide

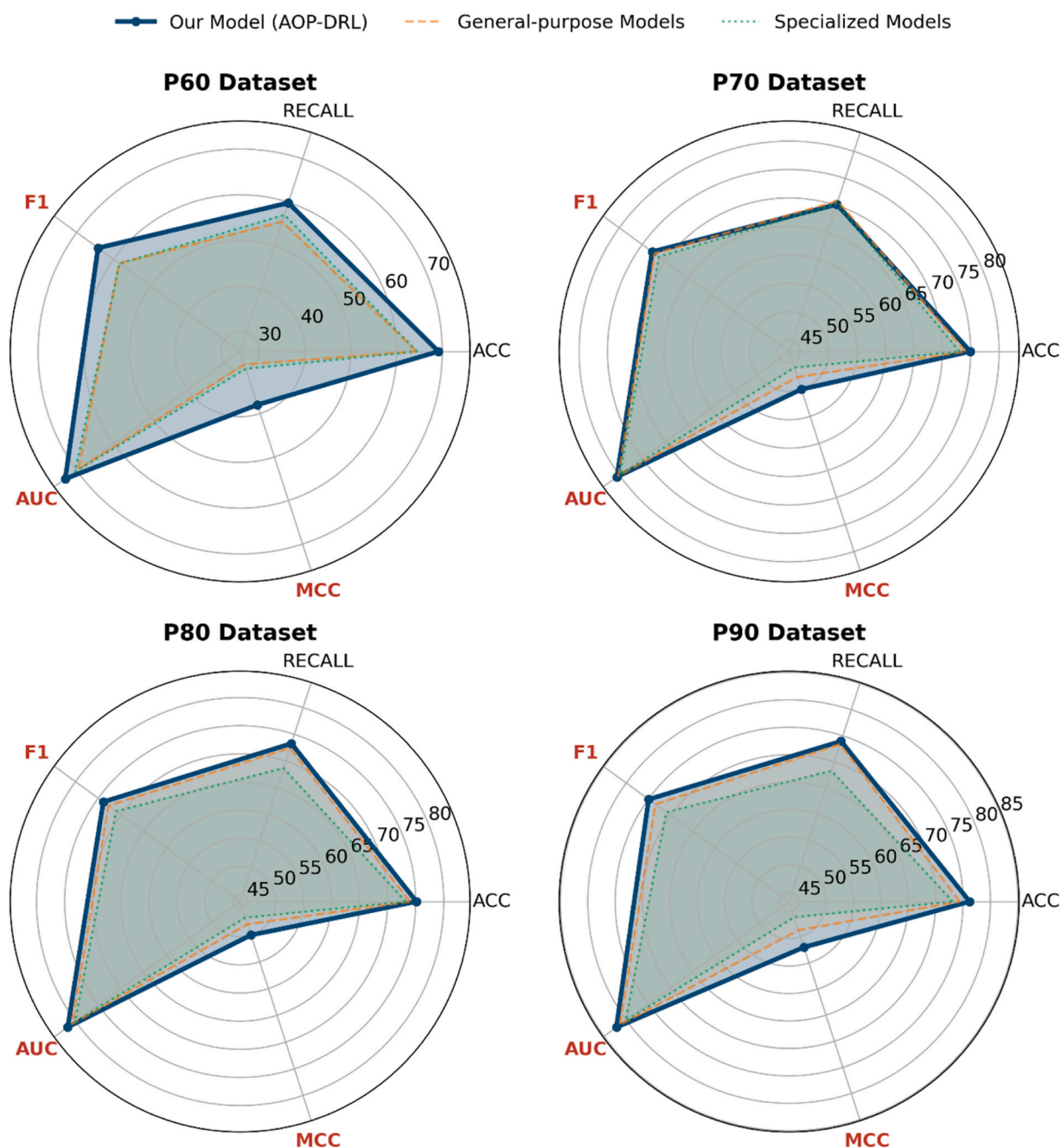


Fig. 6. Comparative performance analysis of AOP-DRL against general and specialized models across multithreshold datasets. Radar charts compare our AOP-DRL model with baseline models on five metrics (ACC/RECALL/F1/AUC/MCC) under four data complexity levels (P60–P90).

properties. These findings align with established principles of protein language model transfer learning [34], confirming that evolutionary-scale pretraining with high-quality data optimally preserves functional conservation patterns.

The fine-tuning layer comparative analysis reveals the distinct advantage of Text-CNN in antioxidant peptide classification. Although LSTM architectures theoretically model long-range dependencies, empirical evidence suggests that antioxidant mechanisms predominantly rely on localized functional motifs, particularly 3–5 residue active site signatures optimally captured by convolutional filters. This observation corroborates recent findings in structural bioinformatics, where Bernhofer *et al.* identified architecture-dependent performance variations across peptide functional classes [35]. Specifically, their transmembrane peptide studies emphasize long-range interaction

modeling, whereas our results demonstrate that antioxidant activity prediction prioritizes short-range physicochemical pattern recognition. This dichotomy highlights how peptide biological function fundamentally dictates optimal computational architecture selection. The ablation study systematically evaluates the contribution of each component of AOP-DRL to the final performance, identifies that the pretrained model plays a central role, and validates the overall design decisions of our framework.

Future methodological developments may benefit from integrating AlphaFold2-derived structural embeddings with multiomics expression profiles across phylogenetically diverse species [36]. Such approaches hold promise for improving the biological plausibility of activity predictions for AOPs [37]. Future development could focus on two conceptual directions: implementing contrastive learning to increase the

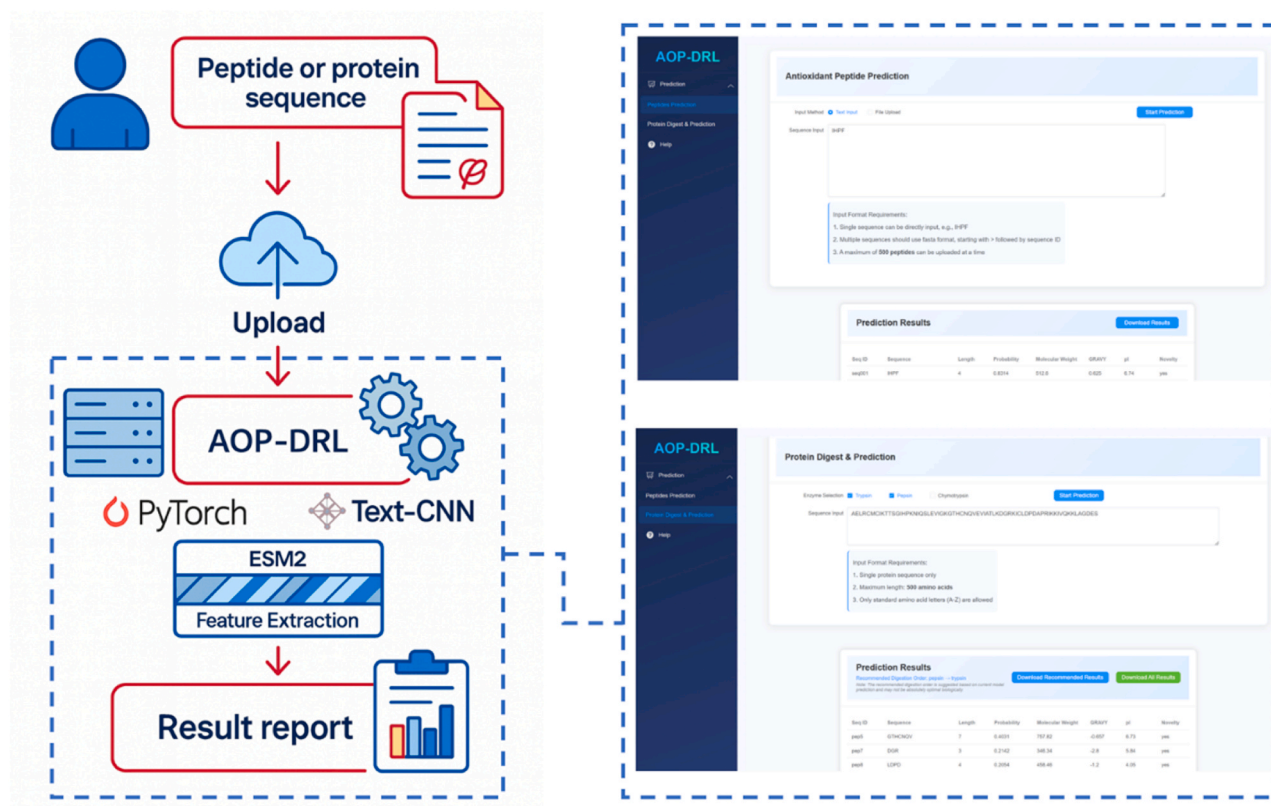


Fig. 7. Operational workflow of AOP-DRL web-server for antioxidant prediction via peptide/protein sequence submissions.

recognition of low-abundance peptides and developing a multimodal integration pipeline incorporating AlphaFold2-derived tertiary structural constraints with expression quantitative trait loci (eQTL) mapping. This dual-axis advancement could bridge the gap between *in silico* predictions and *in vivo* peptide functionality, contributing to more robust computational platforms for precision antioxidant therapeutic development.

In summary, this study establishes the synergistic integration of evolution-scale peptide embeddings provided by ESM-2 and the local feature extraction capability of Text-CNN as the key elements for constructing an efficient AOP-DRL prediction model through systematic ablation analysis. The framework systematically elucidates the structural dependencies governing antioxidant activity efficacy, highlighting its transformative potential in predictive peptide analysis. This provides a high-performance solution for the computational identification of antioxidant peptides. The theoretical foundation and methodological support established in this research will advance the development of precision antioxidant therapy design platforms with enhanced biological relevance and clinical translation potential.

6. Limitations

Although the AOP-DRL framework demonstrates superior classification performance in benchmark evaluations, three primary limitations warrant consideration for translational applications. First, the sensitivity of the model to posttranslational modifications—chemical alterations that modulate antioxidant activity through tertiary/quaternary structural reorganization—remains systematically unvalidated. Second, reliance on *in vitro* biochemical assays introduces potential bias because synthetic peptide systems cannot fully replicate the redox microenvironmental dynamics inherent to physiological contexts (e.g., intracellular glutathione levels and compartment-specific pH gradients). Architecturally, the inductive bias of Text-CNN toward localized sequence motifs inherently limits its capacity to resolve nonlocal

structural determinants of antioxidant function, particularly α -helical arrangements and β -sheet topologies that span membrane interfaces. These geometric constraints directly influence the radical scavenging efficiency through spatial accessibility mechanisms. Future research should validate posttranslational effects, integrate iterative wet-lab validation, and adopt advanced architectures to capture nonlocal structural features for improved antioxidant prediction in complex biological contexts.

7. Conclusions

The AOP-DRL framework synergistically integrates the evolutionary-scale peptide embeddings of ESM-2 with the localized feature extraction capabilities of Text-CNN, systematically establishing the structural dependence of antioxidant efficacy on conserved local motifs. This architecture demonstrated robust generalizability across sequence length variations and low-data regimes. Benchmarking against existing methods revealed consistent superiority across multiple evaluation metrics, highlighting its transformative potential in predictive peptide analytics. Additionally, we have developed and deployed a user-friendly web server, which provides public access to these resources and is expected to greatly benefit and accelerate research in the field of AOPs. Above all, our findings provide a theoretical basis and methodological support for constructing a precision antioxidant therapy design platform with enhanced biological plausibility and clinical translational potential.

CRediT authorship contribution statement

Yongzhu Zhou: Conceptualization, Methodology, Investigation, Writing – original draft, Visualization. **Wanlin Liu:** Writing – review & editing, Formal analysis, Visualization. **Qiao Liu:** Data curation, Validation. **Jie Liu:** Formal analysis, Visualization. **Xing Yu:** Writing – review & editing, Supervision, Project administration, Funding

acquisition. **Jie Ma:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Yunping Zhu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Funding sources

This work was supported by the National Key Research and Development Program of China (2021YFA1301603), National Natural Science Foundation of China (32270697 and 31870828), and the Open Fund of State Key Laboratory of Medical Proteomics (SKLP-O202204).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We thank the National Center for Protein Sciences (Beijing) for computational resources.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.csbj.2025.08.014](https://doi.org/10.1016/j.csbj.2025.08.014).

Data availability

The dataset used in this study is publicly available in the AnOx-PePred repository: <https://github.com/TobiasHeOl/AnOxPePred>. The original source code for this study is publicly available at https://github.com/Zhouyongzhu/AOP_DRL.

References

- [1] Carter R. Oxygen: the molecule that made the world. *J R Soc Med* 2003;96:46–7.
- [2] Lobo V, Patil A, Phatak A, Chandra N. Free radicals, antioxidants and functional foods: impact on human health. *Pharmacogn Rev* 2010;4:118–26.
- [3] DiMartini ET, Lowe CJ, Shreiber DI. Alternative chemistries for free Radical-Initiated targeting and immobilization. *J Funct Biomater* 2023;14.
- [4] Maresca V, Lettieri G, Sorbo S, Piscopo M, Basile A. Biological responses to cadmium stress in liverwort *Conocephalum conicum* (Marchantiales). *Int J Mol Sci* 2020;21.
- [5] Piscopo M, Tenore GC, Notarale R, Maresca V, Maisto M, de Ruberto F, et al. Antimicrobial and antioxidant activity of proteins from feijoa sellowiana berg. Fruit before and after in vitro gastrointestinal digestion. *Nat Prod Res* 2020;34:2607–11.
- [6] Zhu Y, Lao F, Pan X, Wu J. Food Protein-Derived antioxidant peptides: molecular mechanism, stability and bioavailability. *Biomolecules* 2022;12.
- [7] Kupersmidt L, Youdim MBH. The neuroprotective activities of the novel Multi-Target Iron-Chelators in models of alzheimer's disease, amyotrophic lateral sclerosis and aging. *Cells* 2023;12.
- [8] Dreher D, Junod AF. Role of oxygen free radicals in cancer development. *European J Cancer* 1996;32a:30–8.
- [9] Maxwell SR. Coronary artery disease-free radical damage, antioxidant protection and the role of homocysteine. *Basic Res Cardiol* 2000;95(1):165–71.
- [10] Phaniendra A, Jestadi DB, Periyasamy L. Free radicals: properties, sources, targets, and their implication in various diseases. *Ind J Clin Biochem* 2015;30:11–26.
- [11] Ajingi YS, Rukying N, Aroonsri A, Jongruja N. Recombinant active peptides and their therapeutic functions. *Curr Pharm Biotechnol* 2022;23:645–63.
- [12] Neha K, Haider MR, Pathak A, Yar MS. Medicinal prospects of antioxidants: a review. *Eur J Med Chem* 2019;178:687–704.
- [13] Zeng GQ, Zhang PF, Li C, Peng F, Li MY, Xu Y, et al. Comparative proteome analysis of human lung squamous carcinoma using two different methods: two-dimensional gel electrophoresis and iTRAQ analysis. *Technol Cancer Res Treat* 2012;11:395–408.
- [14] Chithrananda S, Grand G., Ramsundar B. ChemBERTa: Large-Scale Self-Supervised Pretraining for Molecular Property Prediction. *arXiv*. 2020;2010.09885.
- [15] Olsen TH, Yesiltas B, Marin FI, Pertseva M, García-Moreno PJ, Gregersen S, et al. AnOxPePred: using deep learning for the prediction of antioxidative properties of peptides. *Sci Rep* 2020;10:21471.
- [16] Lin Z, Akin H, Rao R, Hie B, Zhu Z, Lu W, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* 2023;379:1123–30.
- [17] Du Z, Ding X, Xu Y, Li Y. UniDL4BioPep: a universal deep learning architecture for binary classification in peptide bioactivity. *Brief Bioinf* 2023;24.
- [18] Hesamzadeh P, Seif A, Mahmoudzadeh K, Ganjali Koli M, Mostafazadeh A, Nayeri K, et al. De novo antioxidant peptide design via machine learning and DFT studies. *Sci Rep* 2024;14:6473.
- [19] Elnaggar A, Heinzinger M, Dallago C, Rehawi G, Wang Y, Jones L, et al. ProtTrans: toward understanding the language of life through Self-Supervised learning. *IEEE Trans Pattern Anal Mach Intell* 2022;44:7112–27.
- [20] Zhang N., Bi Z., Liang X., Cheng S., Hong H., Deng S., et al. OntoProtein: Protein Pretraining With Gene Ontology Embedding. *arXiv*. 2022;2201.11147.
- [21] Steinegger M, Mirdita M, Söding J. Protein-level assembly increases protein sequence recovery from metagenomic samples manifold. *Nat Methods* 2019;16:603–6.
- [22] Qin D, Jiao L, Wang R, Zhao Y, Hao Y, Liang G. Prediction of antioxidant peptides using a quantitative structure-activity relationship predictor (AnOxPP) based on bidirectional long short-term memory neural network and interpretable amino acid descriptors. *Comput Biol Med* 2023;154:106591.
- [23] Hassan MT, Tayara H, Chong KT. iAnOxPep: a machine learning model for the identification of Anti-Oxidative peptides using ensemble learning. *IEEE Trans Comput Biol Bioinf* 2025;22:85–96.
- [24] Gimona M. Protein linguistics - a grammar for modular protein assembly? *Nat Rev Mol Cell Biol* 2006;7:68–73.
- [25] Chandra A, Tünnermann L, Löfstedt T, Gratz R. Transformer-based deep learning for predicting protein properties in the life sciences. *eLife* 2023;12.
- [26] Schmirler R, Heinzinger M, Rost B. Fine-tuning protein language models boosts predictions across diverse tasks. *Nat Commun* 2024;15:7407.
- [27] Dickson A, Mofrad MRK. Fine-tuning protein embeddings for functional similarity evaluation. *Bioinformatics* 2024;40.
- [28] He T., Huang W., Qiao Y., Yao J. Text-Attentional convolutional neural network for scene text detection. *IEEE transactions on image processing: a publication of the IEEE Signal Processing Society* 2016. 2525292541.
- [29] Graves A, Schmidhuber J. Framewise phoneme classification with bidirectional LSTM and other neural network architectures. *Neural Netw J Int Neural Netw Soc* 2005;18:602–10.
- [30] Zhang S, Zheng D, Hu X, Yang M. Bidirectional long Short-Term memory networks for relation classification. *Proc 29th Pac Asia Conf Lang Inf Comput* 2015:73–8.
- [31] Xu X, Yang C, Shu K, Yuan X, Li M, Zhu Y, et al. An antibacterial peptides recognition method based on BERT and Text-CNN. *Sheng wu Gong cheng xue bao = Chin J Biotechnol* 2023;39:1815–24.
- [32] Dong G, Wu Y, Huang L, Li F, Zhou F. TExCNN: leveraging pre-trained models to predict gene expression from genomic sequences. *Genes* 2024;15.
- [33] Wang R, Yang X, Pang C, Wei L. A multi-objective comprehensive framework for predicting protein-peptide interactions and binding residues. *IEEE J Biomed Health Inf* 2025.
- [34] Zhu M, Song Y, Yuan Q, Yang Y. Accurately predicting optimal conditions for microorganism proteins through geometric graph learning and language model. *Commun Biol* 2024;7:1709.
- [35] Bernhofer M, Rost B. TMbed: transmembrane proteins predicted through language model embeddings. *BMC Bioinf* 2022;23:326.
- [36] Yang Z, Zeng X, Zhao Y, Chen R. AlphaFold2 and its applications in the fields of biology and Medicine. *Signal Transduct Target Ther* 2023;8:115.
- [37] Kouba P, Kohout P, Haddadi F, Bushuev A, Samusevich R, Sedlar J, et al. Machine Learning-Guided protein engineering. *ACS Catal* 2023;13:13863–95.